Hypervisors, Virtualization & Cloud Networking

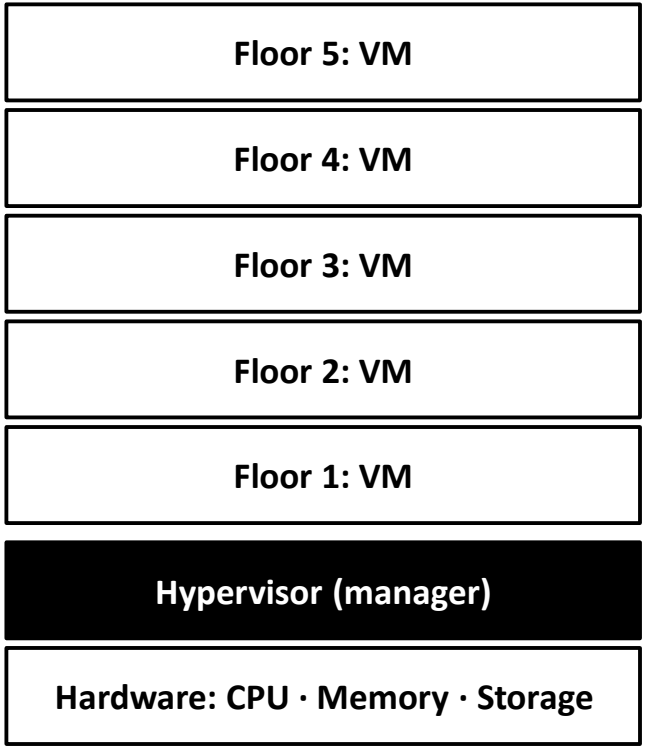
Hypervisor types, the VMware ecosystem, containers, VPC vs VNet, and CIDR

What is a hypervisor

- Software or firmware that creates and manages multiple virtual machines (VMs) on one physical host
- Sits as a middle layer, allocating CPU, memory and storage so each VM runs independently

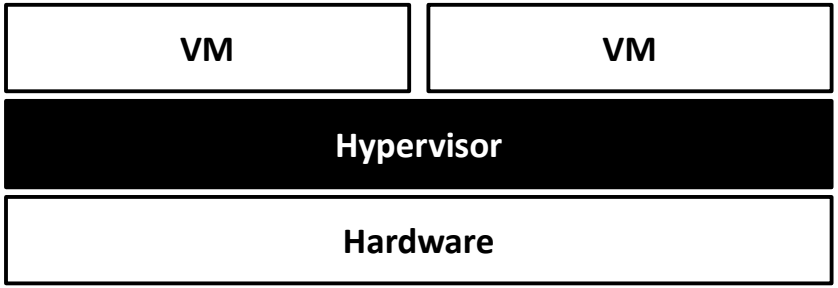
Think of a building:

- Building = physical host
- Floors = VMs
- Building manager = hypervisor
- Electricity, water, network = shared hardware resources



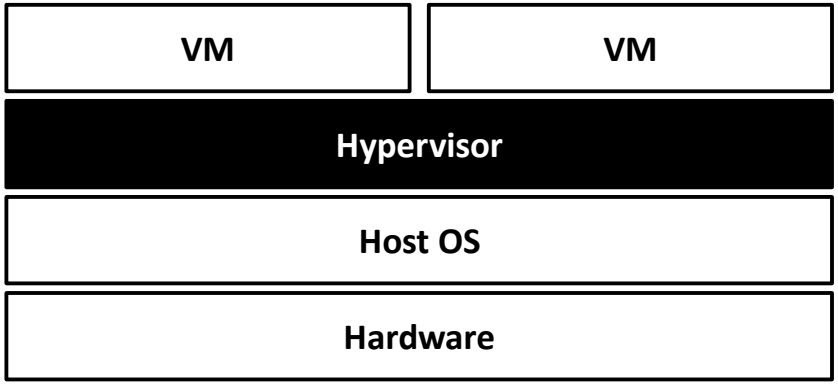
Type 1 vs Type 2 hypervisors

Type 1: Bare-metal



- Runs directly on physical hardware, no host OS
- Used in enterprise data centers for high-performance virtual environments

Type 2: Hosted



- Runs as an application on an existing OS
- Used for development and testing on personal computers

Common hypervisor products

Product	Vendor	Type
VMware ESXi	VMware	Type 1
Microsoft Hyper-V	Microsoft	Type 1
XenServer (Xen)	Citrix	Type 1
Oracle VM	Oracle	Type 1
VMware Workstation	VMware	Type 2
VMware Fusion	VMware	Type 2
Oracle VirtualBox	Oracle	Type 2

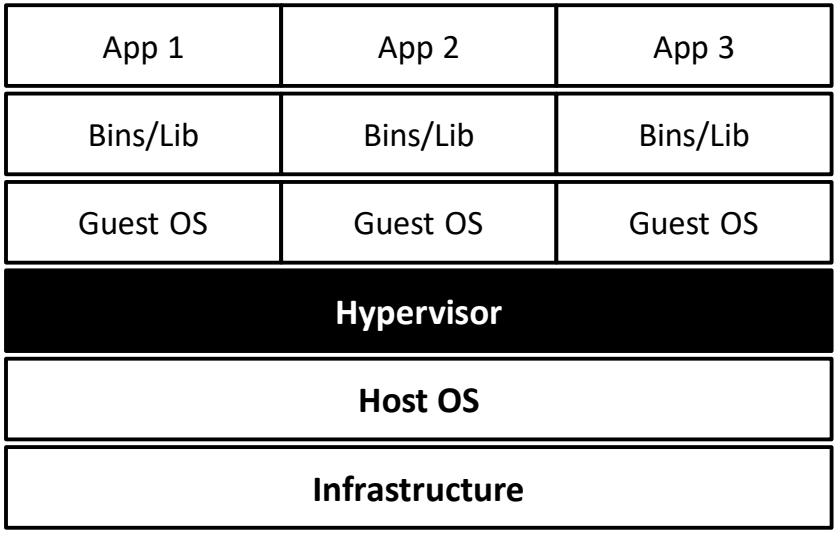
VMware Workstation runs on a Windows or Linux host; VMware Fusion is built for Macs running macOS.

VMware ecosystem

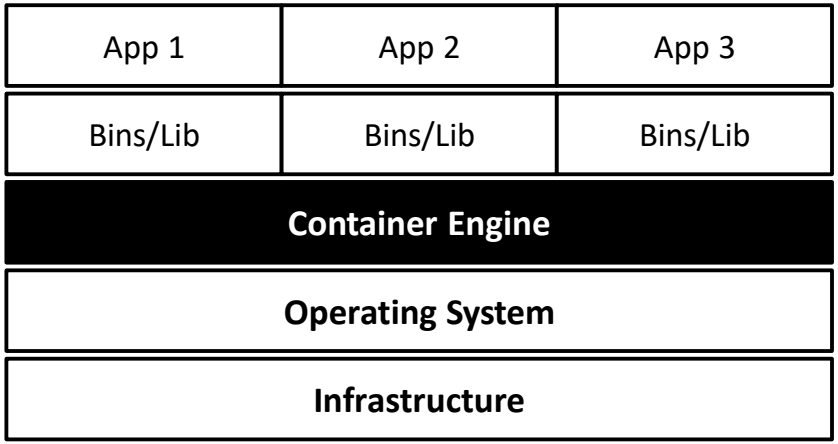
vSphere	Core suite that includes ESXi and vCenter
vCenter Server	Centralized management for multiple ESXi hosts and their VMs
vSAN	Pools disks from multiple ESXi hosts into one shared datastore
NSX	Software-defined networking (SDN): create, manage and secure virtual networks
Horizon	Delivers virtual desktop infrastructure (VDI) and virtual applications to users

Virtualization vs containerization

Virtualization



Containerization



Key difference: each VM carries its own guest OS, while containers share the host operating system.

AWS VPC vs Azure VNet

Feature	AWS VPC	Azure VNet
Purpose	Isolated network to launch resources	Private network for cloud resources
CIDR block	Customizable IP range (e.g., 10.0.0.0/16)	Customizable IP range (e.g., 10.0.0.0/16)
Subnetting	Multiple subnets	Multiple subnets
Internet access	Internet Gateway (IGW), NAT Gateway	Public IP, NAT Gateway
Security	Security Groups, NACLs	Network Security Groups (NSGs), Azure Firewall
Private connections	VPC Peering, AWS Direct Connect	VNet Peering, Azure ExpressRoute
Routing	Route Tables	Route Tables

CIDR (Classless Inter-Domain Routing)

- Method for allocating IP addresses and routing Internet Protocol packets
- Fundamental in the cloud: allocates and manages IP ranges within VPCs, VNets and subnets
- Notation is an IP address, a slash (/), then a number called the prefix length (subnet mask)

